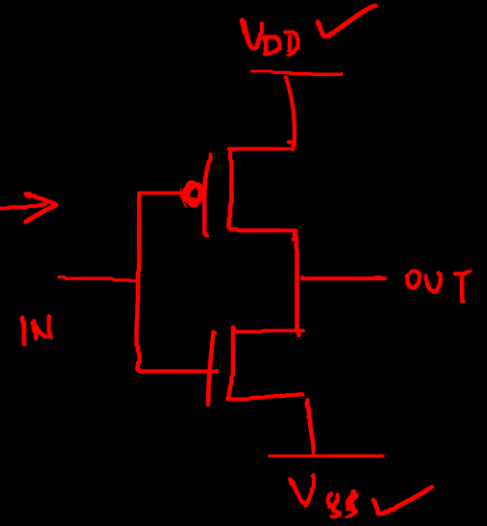


COMPLEMENTARY
MOS.

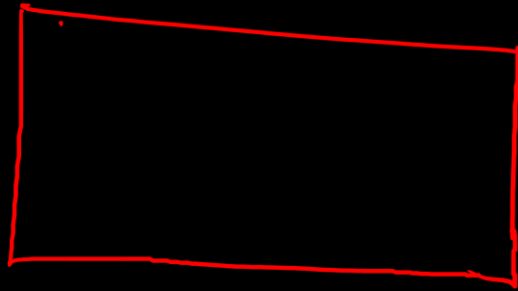


CMOS
realization



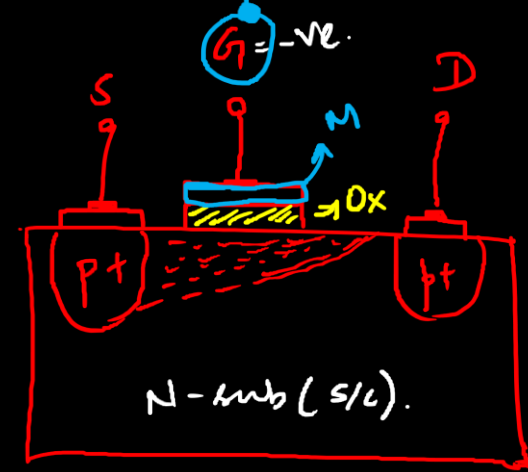
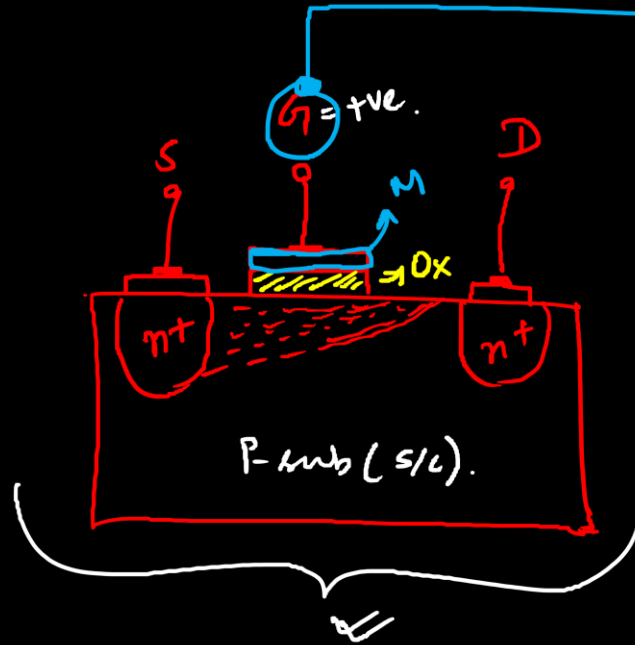
CMOS INVERTER

Er. Sailesh Mishra



N-MOS.

N-MOS $\rightarrow$ N-channel $\rightarrow$ P-sub



P-MOS $\rightarrow$ P-channel $\rightarrow$ N-sub !!

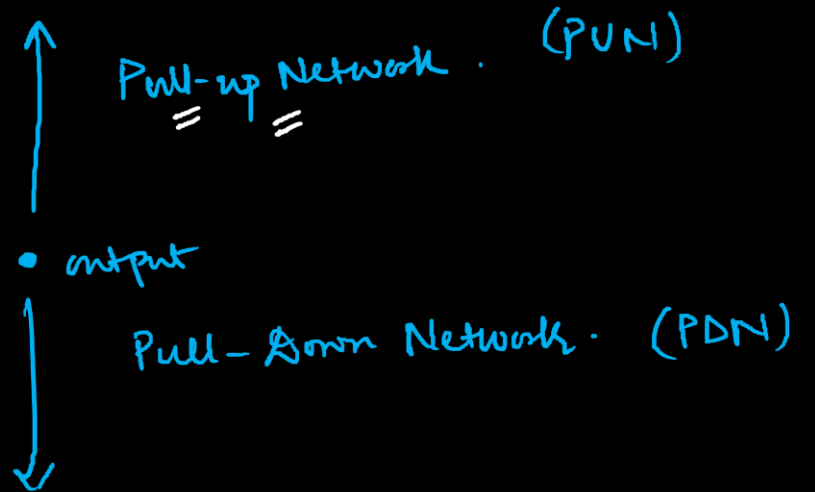
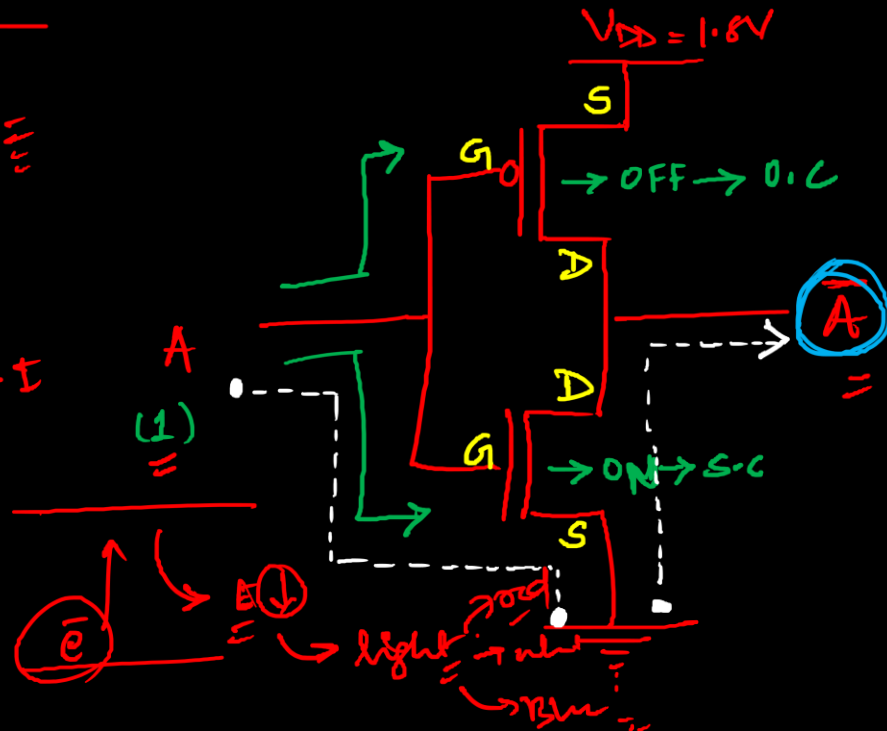
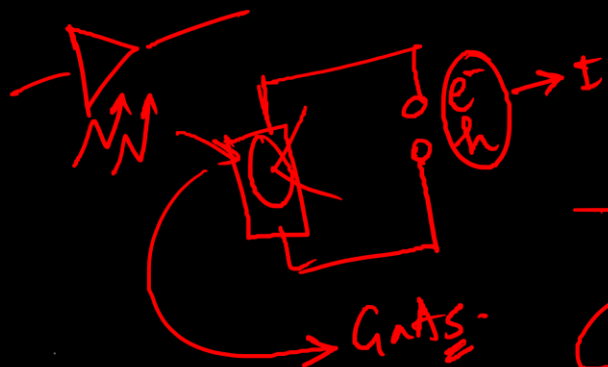
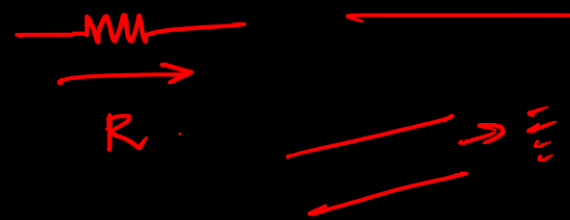
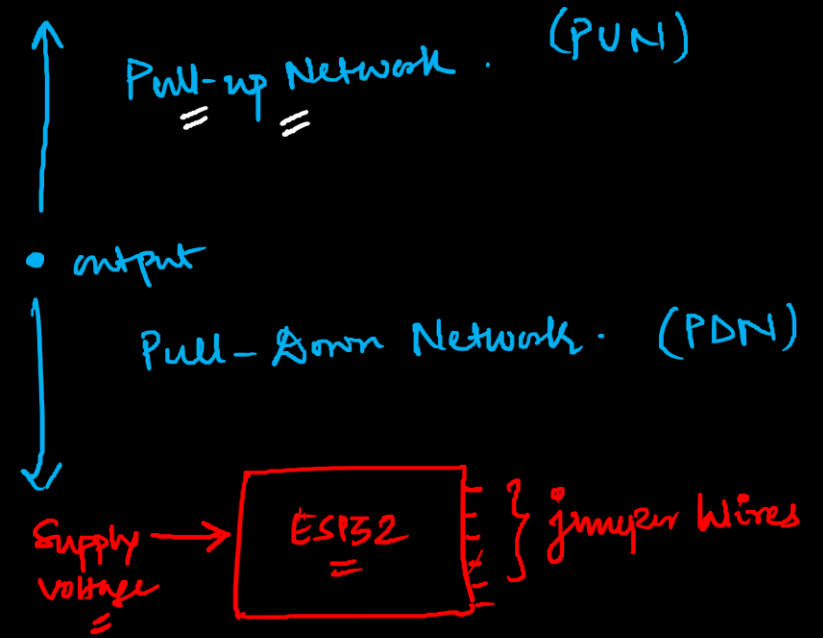
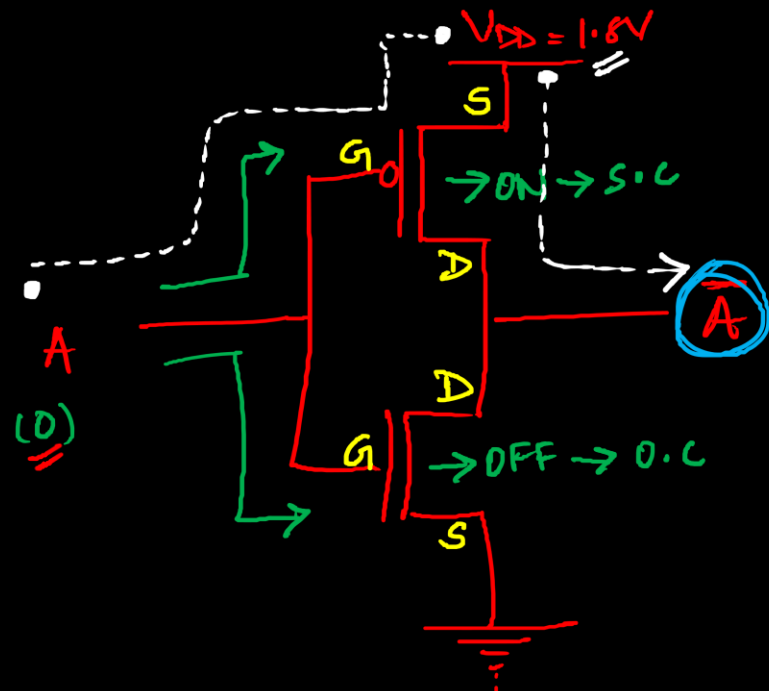
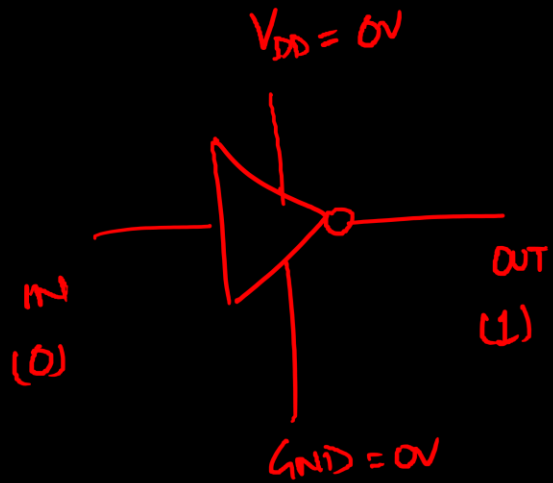
N-MOS $\rightarrow$ P-substrate.

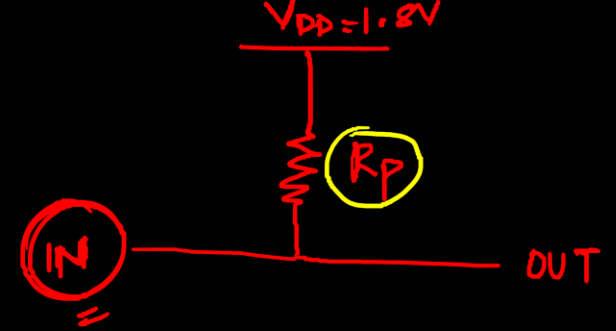
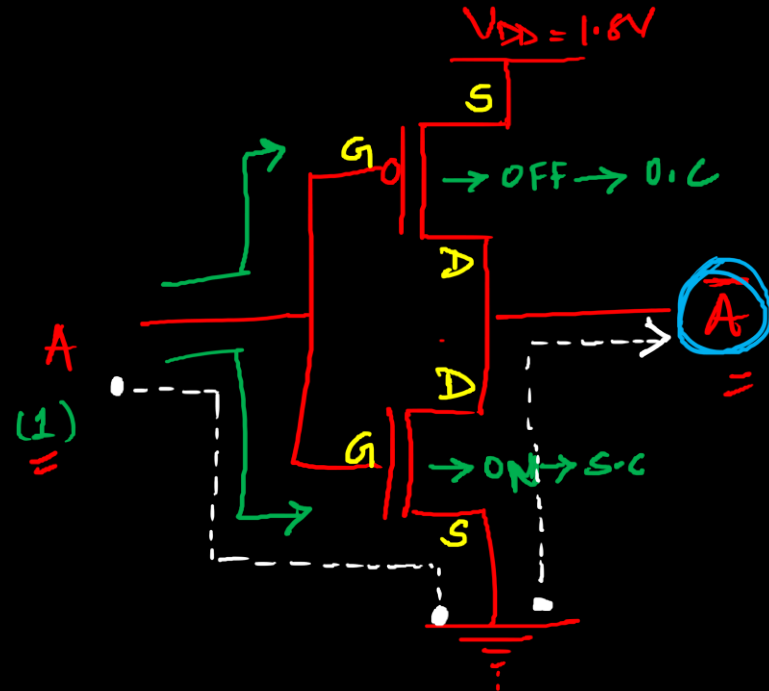
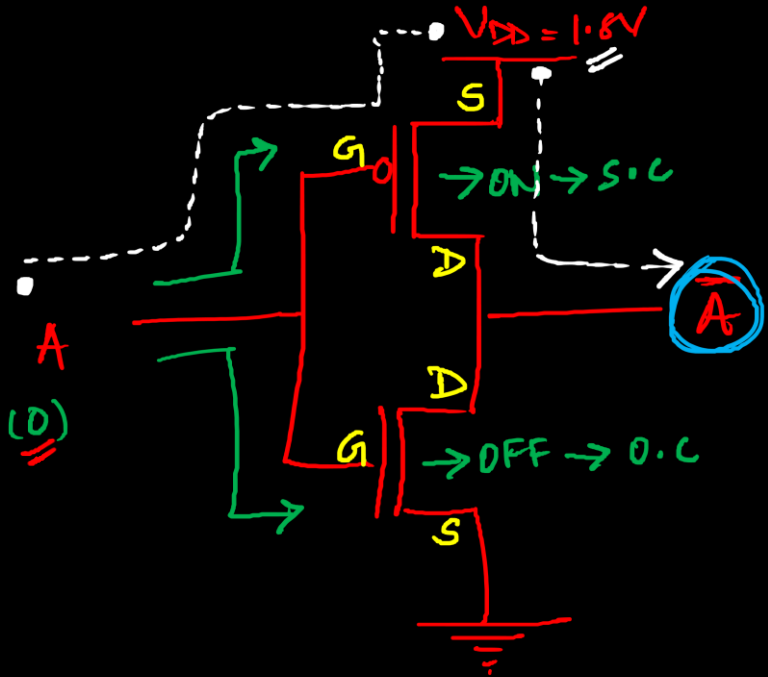
$\rightarrow$ will get TURN ON when Applied '1'

D.c Sweep ($0 \rightarrow 1.8V$) $\rightarrow$ Continuous Time domain.

0 $\rightarrow$ 1 $\rightarrow$ Digital Time domain.

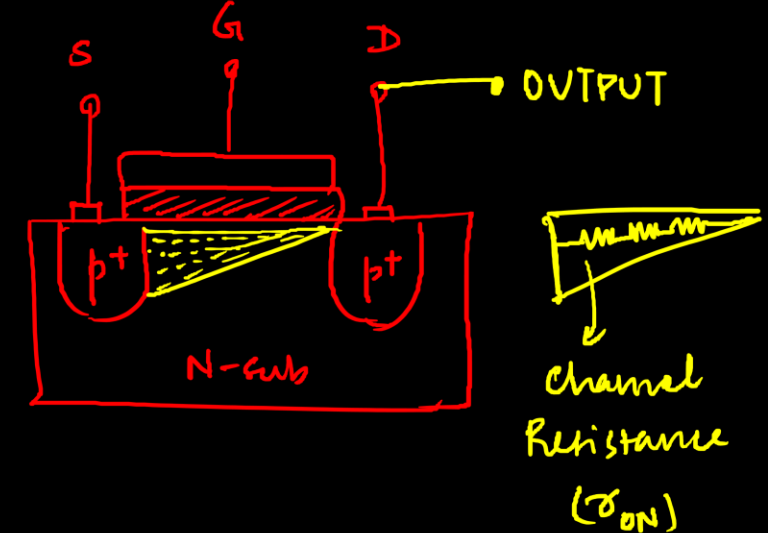
0 1
 $0 \rightarrow 1.8V$: D.c Sweep.



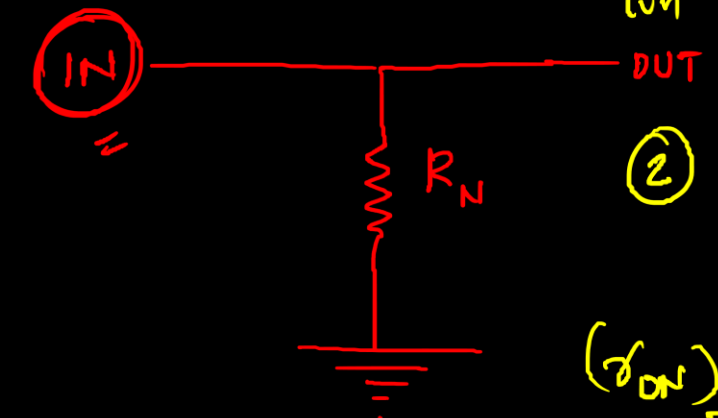


$$i = \frac{V}{R} = \frac{V_{gs}}{r_{onp}}$$

$$\left. \begin{array}{l} i_D \rightarrow \text{lin} \rightarrow \checkmark \\ i_D \rightarrow \text{sat} \rightarrow \checkmark \end{array} \right\} (r_{on})_{Pmos} = ? \checkmark$$



$$(1) (i_D)_{lin} = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \left\{ (V_{gs} - V_{tp}) \cdot V_{ds} - \frac{V_{ds}^2}{2} \right\}$$

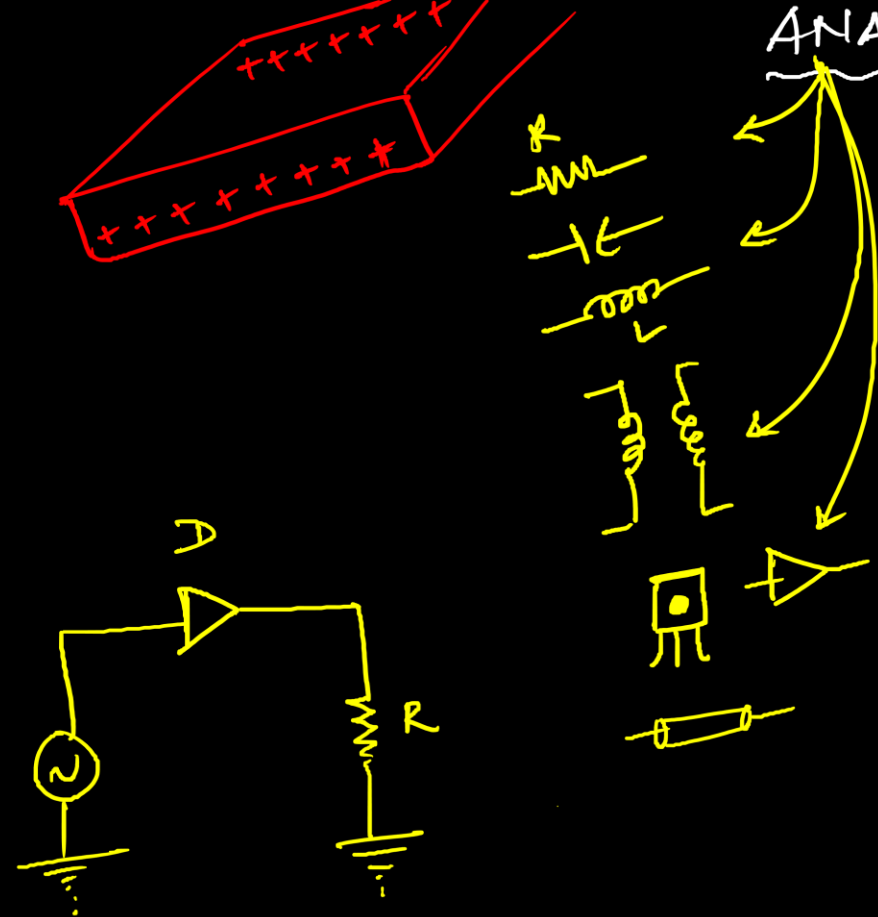


$$(2) (i_D)_{sat} = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \{ V_{gs} - V_{tp} \}^2$$

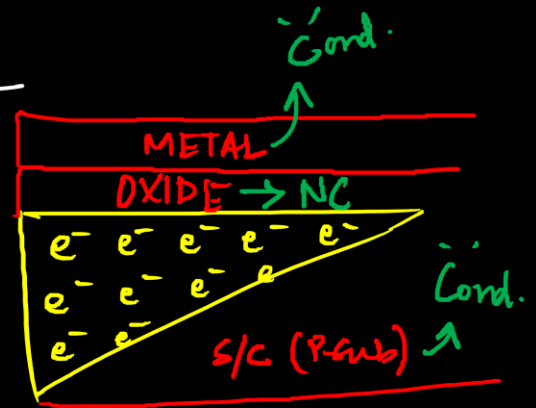
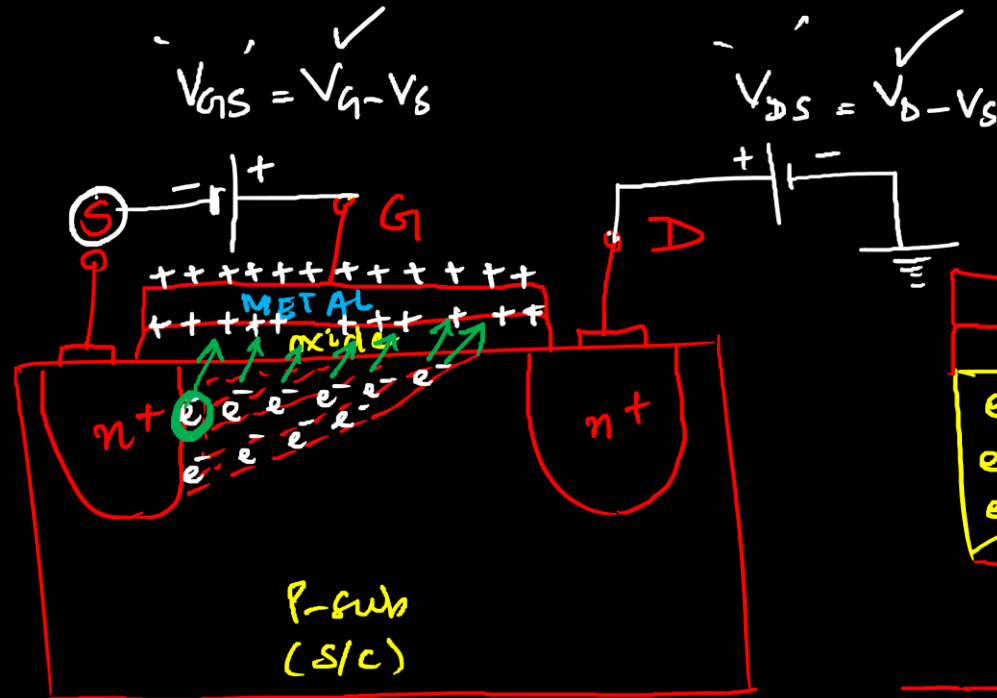
$$(r_{on})_{Pmos} = \frac{V_{gs}}{i_D} = \frac{V_{gs}}{i_D \rightarrow \text{lin} \checkmark \rightarrow \text{sat} \checkmark}$$

ANALOG - CIRCUIT - CHARACTERIZATION

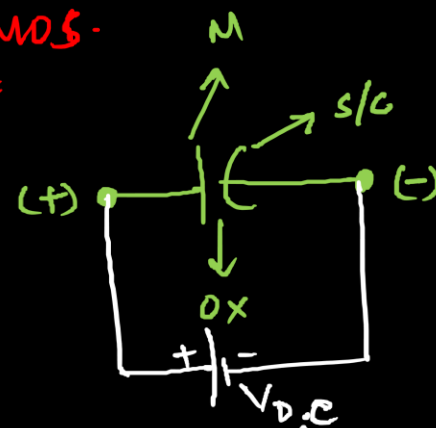
OUTPUT-INPUT ✓



WAVE-Shaping $\frac{0}{\infty}$ [HWR/Clipper etc]
Circuit...

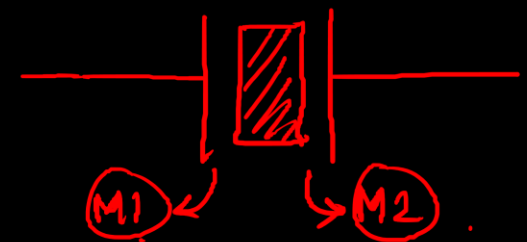


N-MOS.



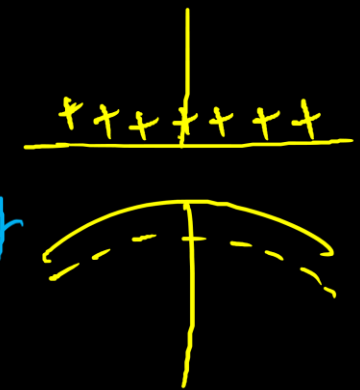
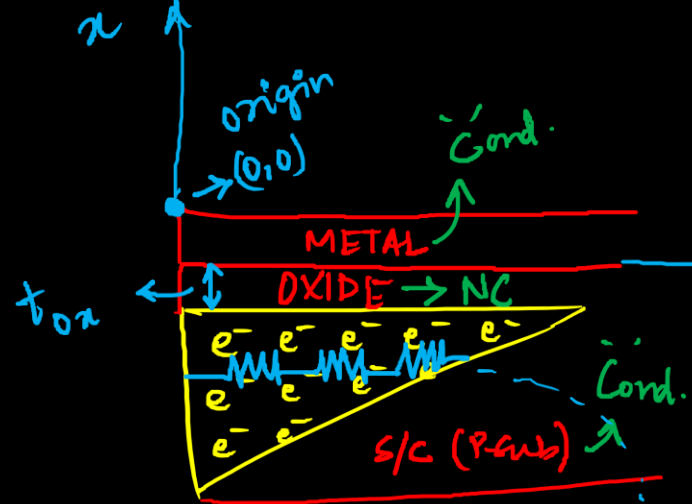
- $\bar{C} (S \rightarrow D)$
- $I_D (S \leftarrow D)$

$\rightarrow$ O/p Current or Drain Current ✓



✓ CAPACITOR ✓

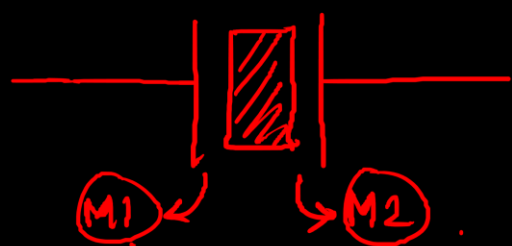
[MOS Device Capacitance]



$$Q = C \cdot V_{ch}(x)$$

$$Q = ne$$

$C_{ox} \equiv$ oxide cap. / ^{per unit} volume
 $= \frac{\epsilon_{ox} \cdot \text{permittivity}}{t_{ox} \cdot \text{thickness of oxide}}$



✓ CAPACITOR
 ↓
 [MOS Device Capacitance]

$(\sigma_{ON}) \rightarrow$ PMOS $\swarrow$
 $\searrow$ NMOS

$\frac{V_{gs}}{I_D} \rightarrow$ $\begin{matrix} \text{lin} \\ \text{Sat} \end{matrix}$

PROCESS PARAMETER

$$\left(\frac{I_D}{V_D} \right)_{lin} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) \left\{ \overbrace{(V_{gs} - V_t)}^{V_{overdrive} \text{ or } V_{effective}} \cdot V_{DS} - \frac{V_{DS}^2}{2} \right\}$$

μ_n → Mobility of e^- inside N-MOS
 $\left(\frac{W}{L} \right)$ → TRANSISTOR'S ASPECT RATIO.
 $(V_{gs} - V_t)$ → CONTROLLING VOLTAGE
 V_{DS} → DRIVING POTENTIAL